

S08 ENHANCED AUDIO DEVICES | O (MAX : 2 PT)

Intent : Improve speech intelligibility and accessibility by providing dedicated, high-performance audio technology.

Summary : This WELL feature requires projects to implement organizational policies and provide occupants with devices that support enhanced speech intelligibility and bolster hearing accessibility in spaces intended for telecommunicating, instruction and public address.

Issue : The ability for people to comprehend speech is a fundamental consideration of universal design. Reduced or low speech intelligibility can negatively impact occupant satisfaction and well-being, especially for non-native speakers, individuals with hearing loss or neurodiverse populations.¹⁻⁹ Audio equipment used for communication can further decrease comprehension of spoken word when installed and used incorrectly.^{1,5,10-13} Additionally, increased auditory workload can impact task performance, resulting in a higher risk of misunderstanding, operational errors and accidents.¹⁴ In educational settings, increased vocal effort by teachers to overcome poor intelligibility has been linked to vocal strain, decreased job performance, lower quality of life, higher rates of leave or absence and resignation.¹⁵⁻²⁰ Improving speech intelligibility can support classroom participation for deaf and hard-of-hearing students, which is linked to improved scores for quality of life, social contact with peers and mental health.²¹

Solutions : Implementing audio systems can improve speech intelligibility for end-users in various environments. These systems include teleconferencing equipment in offices, speech reinforcement systems in classrooms and public address systems.^{22,23} To provide the best possible outcomes for users, systems should be commissioned by a professional in audio engineering in order to meet diverse occupant needs and create more accessible spaces.²⁴⁻²⁶

PART 1 PROVIDE ENHANCED SPEECH INTELLIGIBILITY (MAX : 1 PT)

For All Spaces:

The project meets the following requirements, as applicable:

- a. All rooms that are intended for conferencing, distance learning, or hybrid collaboration contain a combination of microphones, speakers, cameras and supportive audio components (e.g., amplifiers, digital signal processors) that are commissioned by a professional audio engineer.
- b. All public address systems used on a daily basis meet the following:
 1. Commissioned by a professional audio engineer in accordance with NFPA 72 (Annex D), BS 5839 Part 8, ISO 7240 Parts 16 and 19 or equivalent.²⁷
 2. Achieves a minimum STI 0.50 or CIS 0.75 in at least 50% of regularly occupied acoustically distinguishable spaces (ADS) when measured in accordance with IEC 60268-16 or equivalent and as indicated in a commissioning report, acoustical model or similar.²⁸
- c. Speech reinforcement systems are installed in at least 80% of classrooms and all spaces for large presentations (e.g., lecture hall, conference center) and meet the following:
 1. Designed to meet audio distribution requirements in accordance with ANSI/ASA S12.60 Part 1.^{29,30}
 2. Commissioned by a professional acoustician or audio engineer in accordance with ANSI/INFOCOMM A102.01:2017 or equivalent.

PART 2 PRIORITIZE AUDIO DEVICES AND POLICIES (MAX : 1 PT)

For All Spaces:

The project or organization supports individual acoustical needs through policies that meet at least three of the following requirements:

- a. All employer-provided audio devices are managed internally by a qualified professional (e.g., IT professional, mobile device manager) and expectations for use are covered in the employee handbook and/or during on-boarding of new staff.^{31,32}
- b. Eligible employees can request alternative working arrangements to accommodate their individual acoustic comfort needs (e.g., option to work remotely, different workstation location).^{33,34}
- c. Signage is used to indicate both the location and intended activities of the quiet and mixed zones. A minimum of one daily quiet hour is scheduled.³⁵
- d. Eligible employees and distance learners (as applicable) are provided speech-enhancing audio devices (e.g., headsets with active digital signal processing, or noise-cancellation) upon request and at no additional cost or subsidized at least 50%.³⁶

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